

Cloud-Based Data Engineering: An Analysis of Microsoft Azure Solutions

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Abstract — *Cloud computing has become fundamental in data engineering processes as it provides scalable and efficient infrastructure to process large amounts of data. Various cloud-based tools have emerged for data engineering. This paper presents an analysis of cloud-based data engineering, specifically Microsoft Azure. It explores the key Azure services used in data engineering, its strengths, and comparisons with other platforms. In addition, the paper discusses the real-world application of Azure in BMW.*

Keywords— *Cloud computing, data engineering, Microsoft Azure, data pipelines*

EXECUTIVE SUMMARY

This report examines Microsoft Azure in cloud-based data engineering. Microsoft Azure provides a comprehensive set of tools and services that support data engineering processes and other supporting functions.

This report analyses key tools that support different data engineering stages, including Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks and Azure Synapse Analytics. Azure is evaluated based on its strengths, where it is scalable and flexible, cost effective, secure with compliance with data protection requirements. In addition, it is compared with other cloud platforms.

The findings suggest that Azure is a strong and versatile platform for data engineering, especially for integration with other Microsoft products. To benefit from different cloud platforms, a multi-cloud strategy is optimal. The report concludes that cloud-based data engineering is essential for modern organisations, and Azure provides a competitive and scalable solution.

1. INTRODUCTION

Since the explosion of the internet, the volume of data generated has been increasing exponentially. As the terms “data”, “cloud”, “analytics” and “AI” are widely discussed by both tech giants and policymakers, more organisations are acknowledging the importance of data-driven decision making. Data has become a critical asset for modern organisations. As the volume and complexity of data continue to grow, traditional systems struggle with scalability and cost. Consequently, organisations gradually shift towards cloud computing.

The focus of the paper is to examine cloud-based data engineering, specifically Microsoft Azure. This paper aims to analyse Azure tools for data engineering, evaluate its strengths, compare it with other platforms and discuss real-world implications.

2. BACKGROUND

2.1 Data Engineering

According to Reis and Housley (2022), data engineering refers to the development, implementation, and maintenance of processes and systems that transform raw data into consistent and high-quality data for downstream uses, such as machine learning and analytics.

The typical process in data engineering involves the ETL (Extract, Transform, Load) or ELT (Extract, Load, Transform) process. This includes extracting data from various sources or in other terms ingestion, transforming the data with data modeling and structuring, and loading the data into a data warehouse (Crickard, 2020).

With these processes, data engineers are also responsible for the entire infrastructure that the data pipeline runs on. This includes managing and configuring servers, and for those using the cloud, spinning up infrastructure on the cloud platform used (Crickard, 2020).

2.2 Cloud Computing

According to Mell and Grance (2011), “cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.”

This cloud model has the characteristics of on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service, along with three service models, namely Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service (IaaS). The model can be deployed in a private cloud, community cloud, public cloud or hybrid cloud.

Essentially, in cloud computing, everything is hosted in data centers. A data center is a physical facility that houses many servers and other computing equipment. This infrastructure is indispensable in cloud computing as it serves storage, computing power and network connectivity (Andersson, 2023).

2.3 Cloud-based Data Engineering

As organisations increasingly acknowledge the value of data-driven insights, the amount of data generated is scaling exponentially. Traditional data centers cannot meet the demands, and it incurs significant costs. Organisations therefore opt for cloud computing solutions that offer massive parallel processing capabilities and provide strategies to align with an organisation's specific needs (Saminathan, 2024).

There is a wide array of notable cloud services available, such as Amazon Web Services (AWS), Google Cloud Platform (GCP), Azure, Snowflake and Databricks (Reis and Housley, 2022). In this paper, the main focus is Microsoft Azure.

2.4 Microsoft Azure

Microsoft Azure is a computing platform provided by Microsoft. Azure offers a myriad of services, which is divided into several categories in Andersson (2023) including Artificial Intelligence and Machine Learning, Analytics, Compute, Containers, Databases, Developer Tools and DevOps, Integration, Networking, Internet of Things (IoT), Identity and Security, Storage, and Migration and hybrid solutions. This paper focuses specifically on Microsoft Azure in data engineering.

A combination of these services can be used in the data engineering architecture. Figure 1 is an example provided by Riscutia (2021). Data is first ingested into the system and persisted in a storage layer. Then, the transformation phase steps in to aggregate and reshapes the data, which enables analytics and machine learning scenarios. Orchestration and governance are the cross-cutting functions that are paramount across the entire lifecycle. Finally, the data is distributed to the downstream systems.

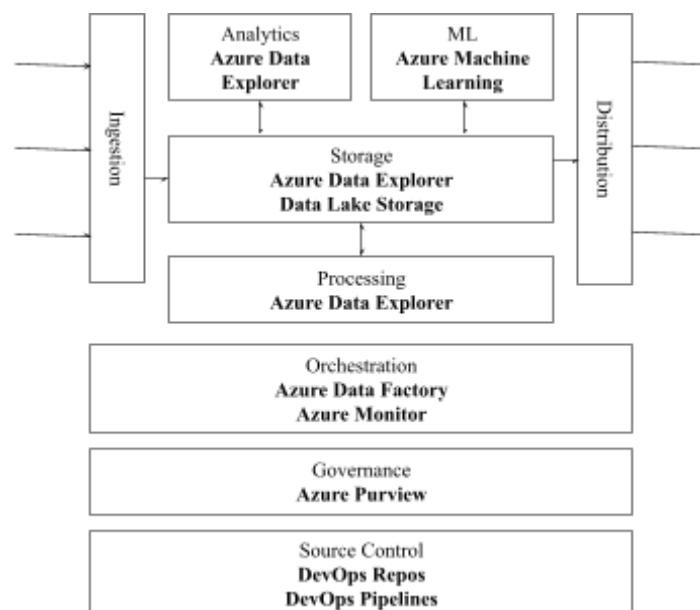


Fig. 1 Architecture with Azure Services (Riscutia, 2021)

3. ANALYSIS AND DISCUSSION

3.1 Azure Capabilities/Tools in Data Engineering

A data engineering pipeline typically includes four stages: data ingestion, data transformation, data storage and data analysis (Behera & Chilukoori, 2024). Microsoft Azure offers a range of integrated services specifically designed to support these stages. This integrated ecosystem simplifies the pipeline development process and improves deployment efficiency.

The key Azure tools that support each stage include:

1) *Azure Data Factory (ADF)*: Azure Data Factory (ADF) is a cloud-based data integration service to design and execute ETL processes (Garudasu et al., 2023). It is critical for modern data engineering as it allows the organisation to move data across various environments. The wide range of data source and destination support data engineers to adapt to different business needs while orchestrating complex workflows.

2) *Azure Data Lake Storage Gen 2*: Azure Data Lake Storage Gen 2 provides a scalable, secure, cloud-based solution for data lake storage to store any size of structured and unstructured data. Organisations can upload a large amount of data into a single repository, with the support of low latency workloads and high performance processing. (Daivi, 2022)

3) *Azure Databricks*: Azure Databricks is an analytics platform that processes large raw datasets using Apache Spark. Databricks can be used to perform basic data pre-processing, such as data cleaning, aggregating and preparing for further analysis. Besides, it supports batch processing, stream processing and machine learning, making it a useful tool especially for big data analytics. (Jadiga, 2024)

4) *Azure Synapse Analytics*: Azure Synapse Analytics integrates data warehousing and analytics services, allowing users to process large scale data. It can be used for end-to-end handling of data, including data ingestion, processing, storage and serving. Besides, through connection with other Azure services like Power BI and Azure Machine Learning, Synapse Analytics is easily extensible with complete analytics solutions. (Jadiga, 2024)

Together, these Azure services form complete data engineering pipelines that are scalable for various situations while reducing the infrastructure complexity and improving the deployment speed.

3.2 Strengths of Azure

There are several advantages of using Microsoft Azure services.

1) *Scalability and Flexibility*: Azure provides a highly scalable and flexible platform for data engineering pipelines. By leveraging Azure services, computing resources can be dynamically allocated depending on the organisation's needs, enabling efficient handling of varying workloads. In addition, Azure is adaptable in diverse use cases where it supports the processing of large volumes of data with different formats and structures. The integrated services environment provides users a one-stop platform for data management without having to invest heavily in physical infrastructure and multiple external services. Furthermore, Azure can be used to absorb and handle large amounts of data regardless of its variety (Singu, 2021). This proves the ability of Azure to scale up or down smoothly.

2) *Cost optimisation*: Azure opted for a pay-as-you-go pricing model. By using this model, the users can pay only when they consume resources and pay only for the amount they consume. This can help to eliminate the problem of capital-intensive investments. Therefore, the pipeline can trade the capital expenses saved for variable expenses. The services under Azure, such as Azure Databricks and Azure Advisor, both have the advantage of auto-scaling and providing personalised suggestions for the resources needed (Singu, 2021). The purpose of these services is to optimise the usage of all Azure resources and to improve the performance of security and reliability (Montazeri, 2025). For instance, when the cost of a single resource can be minimised, it can be used for any other aspect that requires that resource, thus the cost is optimised.

3) *Security and Compliance*: Azure also focuses on enterprise security solutions. It provides a secure environment for data engineers by leveraging Microsoft products, such as Microsoft Defender, Azure Sentinel and Active Directory (Thallam, 2023). In addition to providing services, Azure is always equipped with the latest security like encryption, strict access control measures and compliance with modern data protection requirements (Jadiga, 2024). For example, Azure Security Centre helps organisations to manage and protect their data from being stolen and reduce unauthorised access cases.

Overall, Microsoft Azure provides strong advantages in scalability, cost optimization and security. These strengths make Azure a suitable cloud platform for modern data engineering pipelines that requires efficient data processing and secure data protection.

3.3 Comparison with Other Platforms

The table below shows the key differences between Microsoft Azure and other leading cloud platforms (Amazon Web Services (AWS), Google Cloud Platform (GCP)) :

Table 1. Comparison of Microsoft Azure, Amazon Web Services (AWS) and Google Cloud Platform (GCP)

Feature	Azure	AWS	GCP
Performance and Scalability	Mid-range, strong integration	High performance with strong reliability	Mixed, excels in specific areas
Cost Efficiency	Budget-friendly	Premium pricing	Competitive, discounts
Integration and Ecosystem	Best for Microsoft ecosystems	Versatile across industries	Scalable for data-intensive apps
Use Cases and Strengths	Enterprise integration, cost-saving	Compute-intensive, scalability	Big data, ML, latency-sensitive
Data Processing	Databricks, Synapse	EMR, Glue	Dataflow, Dataproc
Data Warehousing	Synapse	Redshift	BigQuery

1) *Performance*: In terms of performance, Azure provides stable performance. It excels in enterprise integration especially within Microsoft products. According to Nouhas et al. (2025), AWS performs strongly in multiple key criteria like speed, usability and security, making it suitable for organisations that require strong performance and scalability for large-scale workloads. GCP provides mixed performance with less latency bound, making it a good choice for real time big data applications. (Thallam, 2023)

2) *Cost Efficiency*: Azure, AWS and GCP operate on a pay-as-you-go basis. Azure is popular with its cost-effectiveness and the seamless integration with ecosystem engagement (Nouhas et al., 2025). Although AWS is highly flexible with the pay-as-you-go model, it implements a premium pricing strategy especially for smaller-scales projects because of the overhead of combining different services (Moreschini, Hästbacka, & Taibi, 2026). In comparison, GCP is known for its competitive and transparent pricing. It offers sustained use discounts for committed users which can significantly reduce the cost (GeeksforGeeks, 2025).

3) *Integration and Ecosystem*: Azure is excellent in enterprise integration. For organisations that are using Microsoft products, Azure is often the optimal choice as it requires less effort to be integrated and deployed. In contrast, AWS integrates with a wide range of third-party analytics tools, for example, Apache Spark and Tableau, allowing it to easily align with most organisations' needs. GCP is well-suited for data intensive applications as it focuses on advanced capabilities in artificial intelligence and machine learning. (Thallam, 2023)

4) *Use Cases*: Azure is ideal for enterprises leveraging Microsoft products, especially for budget-conscious organisations and customers who previously have a constructive Windows Server License (Hameed et al., 2025). AWS is recognized with its performance, making it suitable for organisations that require compute-intensive and highly scalable applications. GCP is a leader in cutting-edge AI and machine learning, it is more suitable to be used in businesses that rely heavily on big data and artificial intelligence applications.

The comparison between Microsoft Azure, Amazon Web Services and Google Cloud Platform highlights several key differences. Each platform marks different functionalities and unique strengths for different use cases. Azure is ideal for its integration with Microsoft products, AWS is good at its performance and scalability, while GCP's biggest strength is in big data and machine learning applications. Therefore, applying a multi-cloud strategy might be optimal, especially for organisations with diverse requirements.

3.4 Real-World Implications

Microsoft (2024) demonstrates that Azure is used by BMW to reduce analysis times and accelerate prototyping while manually transferring data to on-premises servers. For instance, Microsoft Azure App Service, Azure AI and Azure Kubernetes Service have been used to build an IoT-based Mobile Data Recorder (MDR) system for BMW. These tools significantly increase analysis speed. In terms of data storage and warehousing, BMW utilises Azure Data Lake and Blob Storage to manage structured and unstructured vehicles data captured (Thallam, 2023) and Azure Synapse Analytics serves as the centralised data warehouse. This cloud-based approach improved speed of analysis and reduced the time for data processing. Additionally, Azure services

such as Azure Data Explorer and Azure Databricks have been utilised to deliver a faster and more efficient data analysis. This shows that Azure is highly suitable for large-scale data and real-time processing applications.

4. RECOMMENDATIONS

4.1 Choosing the Right Cloud Provider

Based on the findings from this study, no single cloud platform can be considered universally optimal for all organisations. Therefore, selecting a cloud platform for an organisation is not about choosing the best one, but we need to align the platform with its business requirements and workloads. The most suitable scenarios for each cloud platform are summarised in the decision table below.

Table 2. Recommended Cloud Platforms for Different Use Cases

Scenario	Recommended Platform
Microsoft-based company	Microsoft Azure
High-performance computing	Amazon Web Services
AI / Machine Learning projects	Google Cloud Platform

4.2 Future Trend and Adoption

As data engineering continues to grow, more organizations adopt cloud platforms to process large-scale data and perform real-time data analysis. Microsoft Azure provides tools such as Azure Data Factory and Azure Databricks to support the entire data pipeline, from data ingestion to analysis. Therefore, it is important for the organisation to consider both current needs and future scalability before choosing a cloud platform.

CONCLUSION

Cloud-based data engineering is becoming the preferred choice for modern organizations to process the increasing amount of data. In this paper, the Azure solution is examined. With its robust suite of services, it can be adopted to meet the diverse business needs of different organisations. This includes Azure Data Factory that handles the ETL processes, Azure Data Lake Storage Gen 2 that offers scalable storage, Azure Databricks that performs data preprocessing, and Azure Synapse Analytics that integrates data warehousing and analytics. Azure has become a preferred option due to its scalability and flexibility, pay-as-you-go payment model that allows cost optimisation and its compliance with security requirements. Despite its strength, Azure is just one of the several platforms. There are many competitors, including tech giants like Amazon Web Services and Google Cloud Platform. Each serves their own unique strengths and functionalities. An organisation should evaluate the needs and ensure that the curated platform aligns with business needs. In conclusion, cloud computing is not just a trend but a must, especially for large organisations dealing with vast amounts of data. Azure is among the competitive cloud-platforms that should be considered.

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APPENDIX

Name: Goe, Lam Datetime: 1/4/2026 18:45 - 18:55	Lam: Executive Summary, Introduction, Background, Conclusion Goe: Analysis and Discussion, Recommendation
Name: Goe Datetime: 3/4/2026 11:15 - 11:33	Task: Determine each subsection in Analysis and Discussion Section
Name: Goe Datetime: 3/4/2026 15:50 - 16:45	Task: Using Google Scholar and scopus to do research for related materials Prompt Keyword: “Azure in data engineering”, “Challenge of Azure in data engineering”, “Comparison between Azure and other platforms”
Name: Goe Datetime: 5/4/2026 00:58 - 02:09	Task: Draft some advantages and comparison with other platforms Reference: 1. Singu, S. K. (2021). Designing scalable data engineering pipelines using Azure and Databricks. <i>ESP Journal of Engineering & Technology Advancements</i>, 1(2), 176-187. 2. Montazeri, K. (2025). <i>Cost Optimization in Azure Data Engineering</i>. 3. Sridhar, R., Dhenia, R. N. K., & Kanani, I. J. (2023). The Impact of Cloud Computing on Data Engineering: Leveraging AWS, Google Cloud, and Azure. <i>International Journal of Emerging Trends in Computer Science and Information Technology</i>, 4(3), 171-184.
Name: Goe Datetime: 5/4/2026 15:24 - 16:59	Task: Complete advantages section and draft the tools Reference: 1. Jadiga, S. (2024). Big Data Engineering Using Hadoop and Cloud (GCP/AZURE) Technologies. <i>International Journal of Computer Trends and Technology</i>, 72(8), 60-69. 2. Behera, L., & Chilukoori, V. V. R. (2024). End-to-end data pipelines: Redefining the architecture of data engineering in cloud environments. <i>ESP International Journal of Advancements in Science & Technology</i>, 2(4), 26-33.
Name: Goe Datetime: 6/4/2026 21:10 - 23:13	Task: complete the Azure tools Prompt: “Match all the tools with the four stage of data engineering pipeline” Reference: 1. Garudasu, S., Ganipaneni, S., Kshirsagar, R. P., Goel, O., Jain, A., & Goel, P. (2023). Scalable ETL processes in Azure Data Factory: Best practices

	for data engineers. <i>International Journal of Research in Modern Engineering and Emerging Technology (IJRMEET)</i> , 11(4).
Name: Lam Datetime: 6/4/2026 16:33 - 16:59	Task: Using Google Scholar to do research Keyword Used: - Cloud-Based Data Engineering: An Analysis of Microsoft Azure Solutions - Cloud-Based Data Engineering with Microsoft Azure
Name: Lam Datetime: 8/4/2026 11:43-14:30	Task: Using Google Scholar to do research and skim through articles - Thallam, N. S. T. (2023). Comparative Analysis of Public Cloud Providers for Big Data Analytics: AWS, Azure, and Google Cloud. <i>International Journal of AI BigData Computational and Management Studies</i>, 4, 18–29. https://doi.org/10.63282/3050-9416.ijaibdcms-v4i3p103 - Sridhar, R., Dhenia, R. N. K., & Kanani, I. J. (2023). The impact of cloud computing on data engineering: leveraging AWS, Google Cloud, and Azure. <i>International Journal of Emerging Trends in Computer Science and Information Technology</i>, 4, 171–184. https://doi.org/10.63282/3050-9246.ijetcsit-v4i3p118 - Stendal, K., & Westin, S. (2018). Implementing Cloud Based Big Data Platforms – A Case Using Microsoft Azure. <i>Duo Research Archive (University of Oslo)</i>, 26(1). http://hdl.handle.net/11250/2596543 - Chowdhury, R. H. (2020). Cloud-Based Data Engineering for Scalable Business Analytics solutions: Designing scalable cloud architectures to enhance the efficiency of big data analytics in enterprise settings. <i>Journal Technological Science & Engineering (JTSE)</i>, 2(1). https://doi.org/10.5281/zenodo.15043092 Task: Define structure of “2. Background” section Task: Using Google Scholar and Scopus to do research and skim through articles, some are also searched directly in the browser Keywords Used: “Data Engineering”, “Data Engineering process” - Reis, J., & Housley, M. (2022). <i>Fundamentals of Data Engineering: Plan and Build Robust Data Systems</i>. “O’Reilly Media, Inc.” Keywords Used: “Cloud Computing” - Jamsa, K. (2022). <i>Cloud computing</i>. Jones & Bartlett Learning. - Microsoft. (n.d.). <i>What is cloud computing?</i> https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-cloud-computing/?msocid=2fe52df6985d62381d513ac2998163ed Engineering Data Processing Workflows IEEE Journals & Magazine IEEE Xplore
Name: Goe Datetime: 8/4/2026 12:10 - 13:09	Task: Review previous article found and conclude the differences in a comparison table References: - Nouhas, H., Belangour, A., & Nassar, M. (2025). A weighted scoring model analysis of cloud storage services: Comparing AWS, Azure, and Google Cloud Platform. <i>International Journal of Engineering Trends and Technology</i>, 73(6), 350–370. https://doi.org/10.14445/22315381/IJETT-V73I6P129 - Hameed, S., Bukhari, S. H. R., Rizvi, S. U. E. A., Tahir, M., & Aadil, F. (2025). A comparative analysis on the services offered by Amazon Web Services and Microsoft Azure. <i>Concurrency and Computation: Practice and Experience</i>. https://doi.org/10.1002/cpe.70215 - Moreschini, S., Hästbacka, D., & Taibi, D. (2026). <i>Comparing cloud-native MLOps platforms: AWS, Azure, GCP, and Databricks</i>. <i>IEEE Software</i>, 43(2), 136–142.

	https://doi.org/10.1109/MS.2025.3646505
Name: Lam Datetime:	Task: Working on Background - Data Engineering 1. Crickard, P. (2020). <i>Data Engineering with Python: Work with massive datasets to design data models and automate data pipelines using Python</i> . Packt Publishing Ltd.
Name: Goe Datetime: 10/04/2026 21:07 - 21:37	Study the real world applications of Azure (BMW) References: 1. Microsoft. (2025, January 28). BMW Group innovates with Azure for 10 times more efficient data delivery. https://www.microsoft.com/en/customers/story/19769-bmw-ag-azure-app-servive 2. Thallam, N. S. T. (2023). Comparative Analysis of Public Cloud Providers for Big Data Analytics: AWS, Azure, and Google Cloud. <i>International Journal of AI BigData Computational and Management Studies</i> , 4, 18–29. https://doi.org/10.63282/3050-9416.ijaibdcms-v4i3p103
Name: Lam Datetime: 10/04/2026	Task: Working on Background Keyword: Cloud Computing, Cloud-based Data Engineering, Azure 1. Mell, P. M., & Grance, T. (2011). <i>The NIST definition of cloud computing</i> . https://doi.org/10.6028/nist.sp.800-145 2. Gopa, R., & Roy, A. (n.d.). <i>Data Engineering on the Cloud: A Practical Guide 2025</i> . YASHITA PRAKASHAN PRIVATE LIMITED. 3. Saminathan, M. (2024). <i>Mastering Big Data Engineering: AWS, GCP, & Azure Showdown</i> . Libertatem Media Private Limited. 4. Andersson, J. C. (2023). <i>Learning Microsoft Azure</i> . “O’Reilly Media, Inc.” Working on Abstract, Executive Summary, Introduction
Name: Goe Datetime: 11/04/2026	Task: Write the recommendations Prompt: how to write a recommendation section in a technical report?
Name: Lam Datetime: 12/04/2026	Task: Write conclusion Revise entire document